

Chapter 6: Reactions & Equilibrium

Acids, Bases, Redox Reactions, Chemical Kinetics & Le Chatelier's Principle

6.1 Acids, Bases & pH Calculations

- **Arrhenius Theory:** Acids produce H^+ ; bases produce OH^- .
- **Brønsted-Lowry Theory:** Acids are proton donors; bases are proton acceptors.
- **Lewis Theory:** Acids are electron-pair acceptors; bases are electron-pair donors.
- $\text{pH} = -\log[\text{H}^+]$, with $\text{pH} + \text{pOH} = 14$ at standard conditions.

6.2 Chemical Kinetics & Reaction Rates

Kinetics studies the speed of reactions and their mechanisms. Rate laws relate reaction rate to reactant concentrations: $\text{Rate} = k[\text{A}]^x[\text{B}]^y$. Catalysts lower activation energy (E_a) without being consumed in the reaction.

6.3 Chemical Equilibrium

At equilibrium, forward and reverse reaction rates are equal. The equilibrium constant expression for $a\text{A} + b\text{B} \rightleftharpoons c\text{C} + d\text{D}$ is:

$$K_c = \frac{[\text{C}]^c[\text{D}]^d}{[\text{A}]^a[\text{B}]^b}$$

Le Chatelier's Principle: If a dynamic equilibrium is disturbed by changing conditions, the position of equilibrium shifts to counteract the change.